CHE 565 -- Homework 5

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Problem 1

Closed-loop Response to Step Disturbance

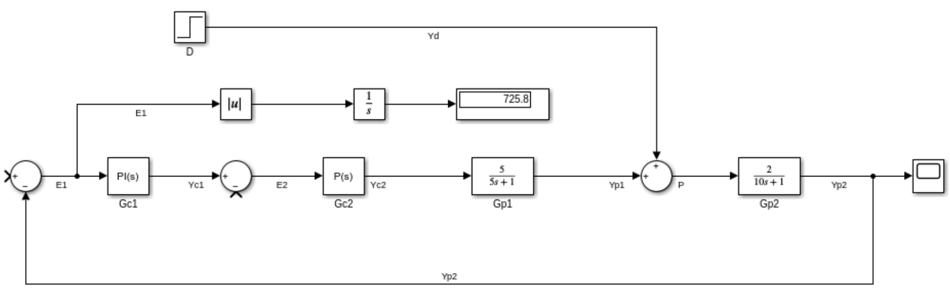

Figure 1: Block diagram of the closed-loop system without cascade control and with a unit step disturbance from Simulink.

The Simulink block diagram is shown in Figure 1.

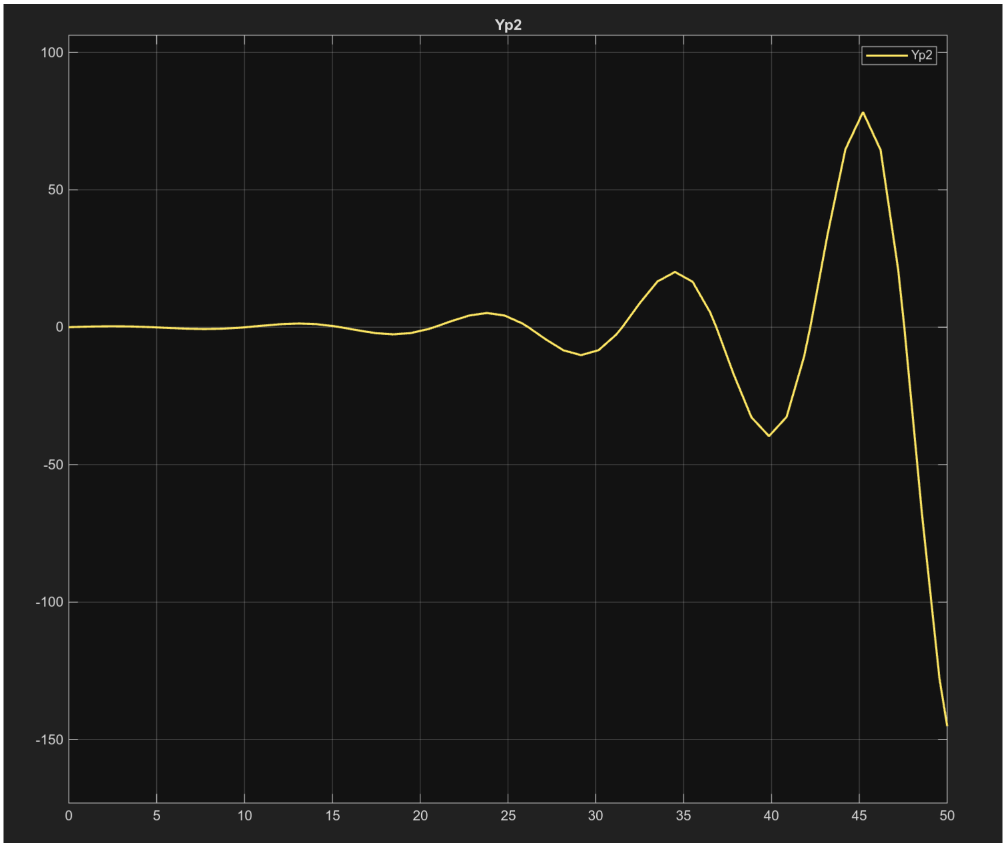

Figure 2: Closed-loop response to no step change in setpoint and a unit step change in disturbance from Simulink.

The Simulink closed-loop disturbance response is shown in Figure 2.

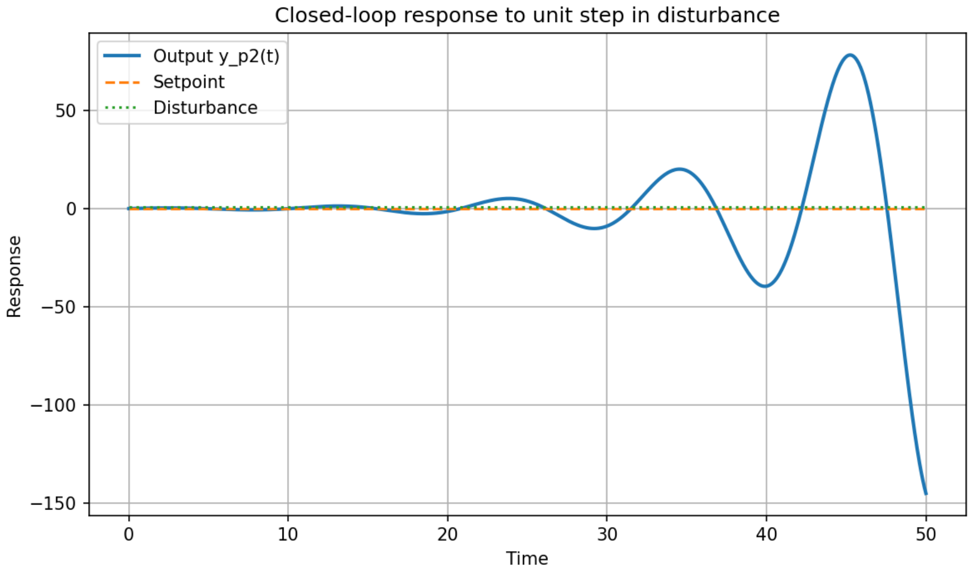

Figure 3: Closed-loop response to no step change in setpoint and a unit step change in disturbance from Python.

The corresponding Python response is shown in Figure 3.

Case 1: Step disturbance with no setpoint change gave IAE = 725.3696 in Python and IAE = 725.8109 in Simulink.

The PI controller was then tuned using the autotuning feature in Simulink, which gave P = 0.1837 and I = 0.0816.

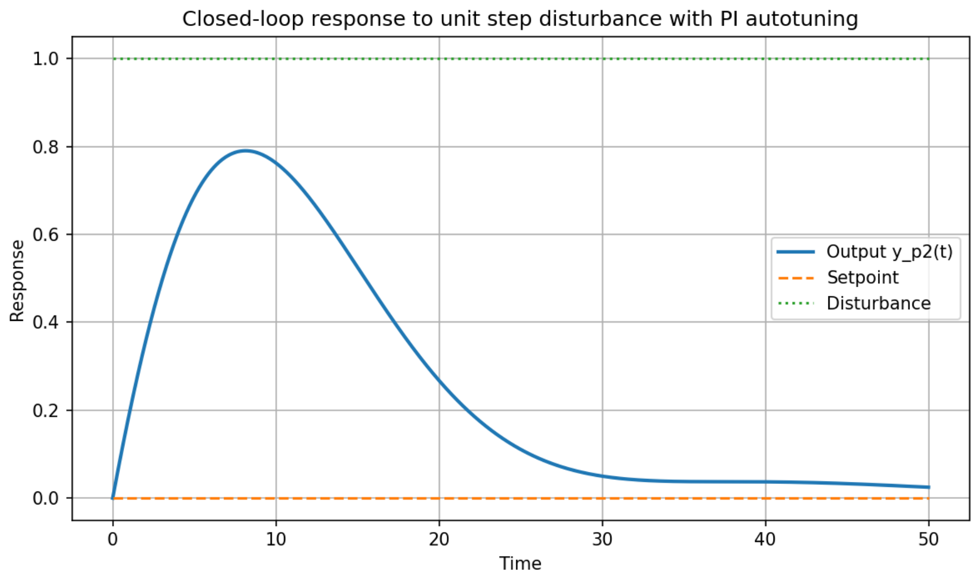

Figure 4: Closed-loop response to no step change in setpoint and a unit step disturbance using the Simulink autotuned PI values in Python.

The Python response using the autotuned PI values is shown in Figure 4.

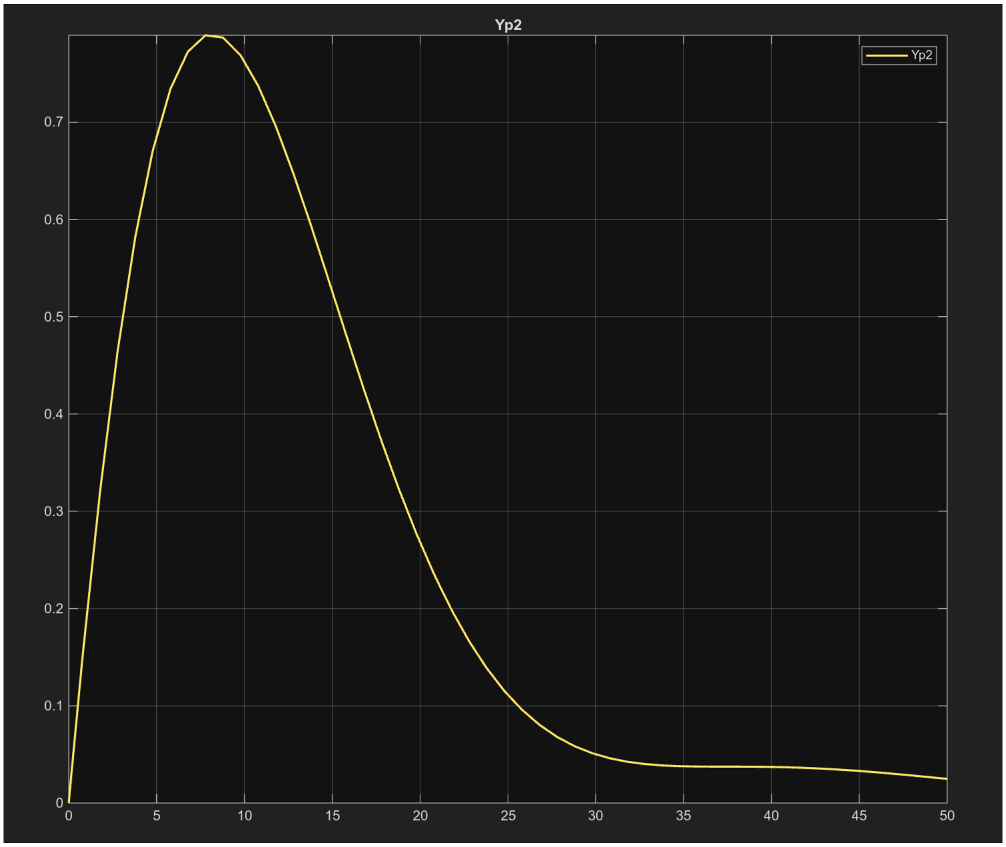

Figure 5: Closed-loop response to no step change in setpoint and a unit step disturbance using PI autotuning from Simulink.

The corresponding Simulink response using PI autotuning is shown in Figure 5.

Case 2: Step disturbance with no setpoint change and PI autotuning gave IAE = 13.0462 in Python and IAE = 13.0463 in Simulink.

Closed-loop Response to Step Setpoint Change

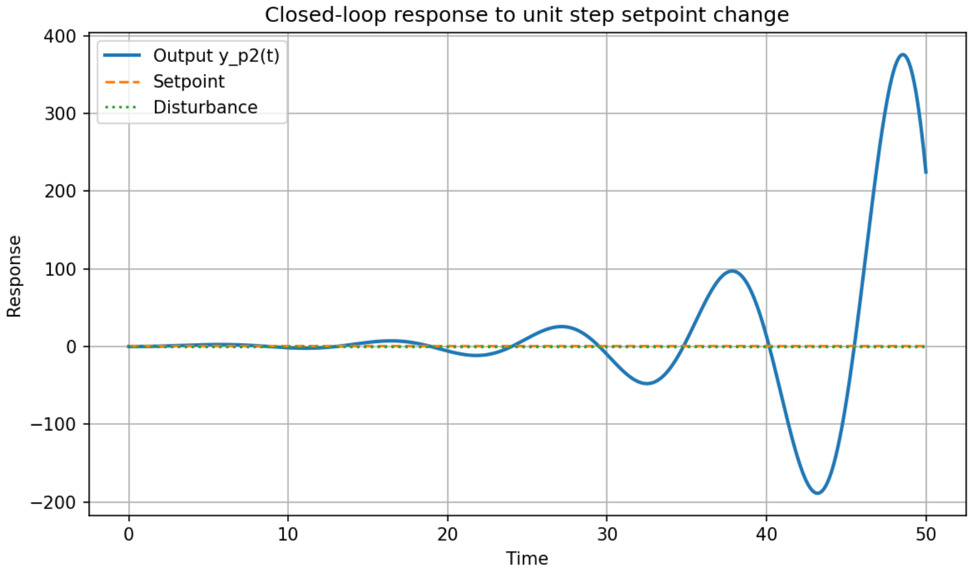

Figure 6: Closed-loop response to unit step setpoint change with no disturbance change from Python.

Next, the response to a unit step change in setpoint with no disturbance change was simulated. The Python response is shown in Figure 6.

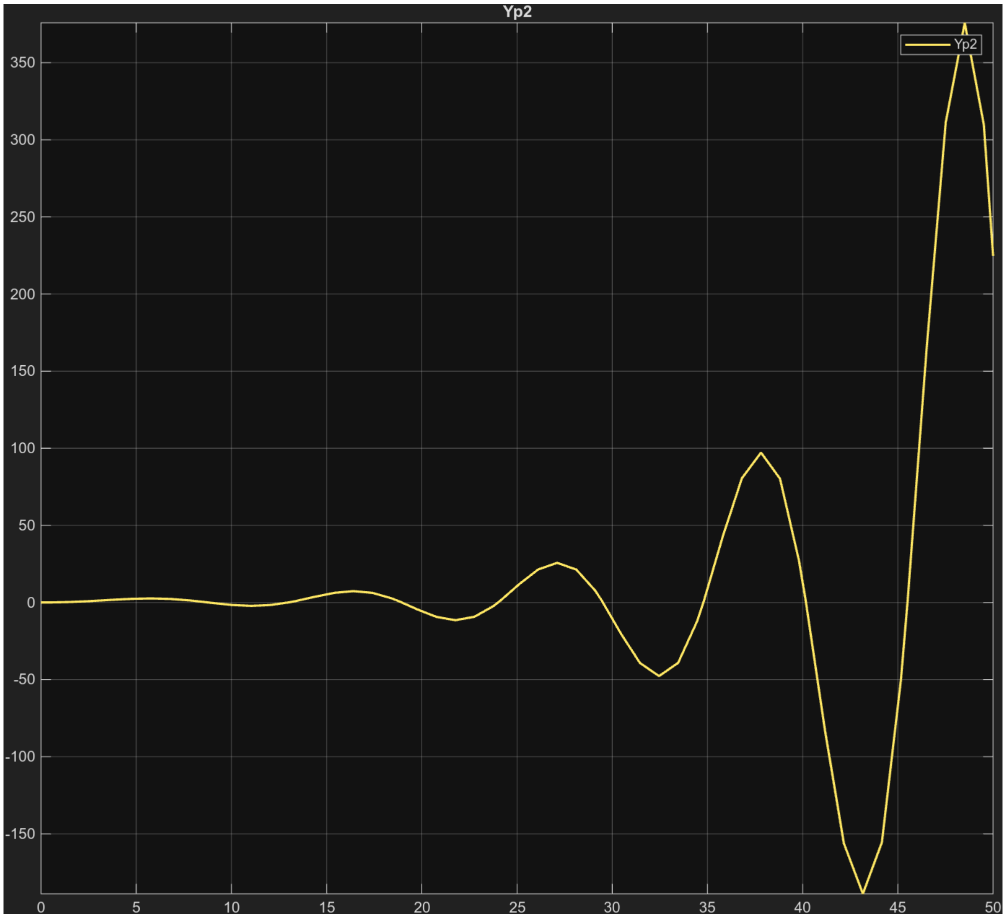

Figure 7: Closed-loop response to unit step setpoint change with no disturbance change from Simulink.

Case 3: Step setpoint change with no disturbance change gave IAE = 2449.4065 in Python and IAE = 2450.8205 in Simulink.

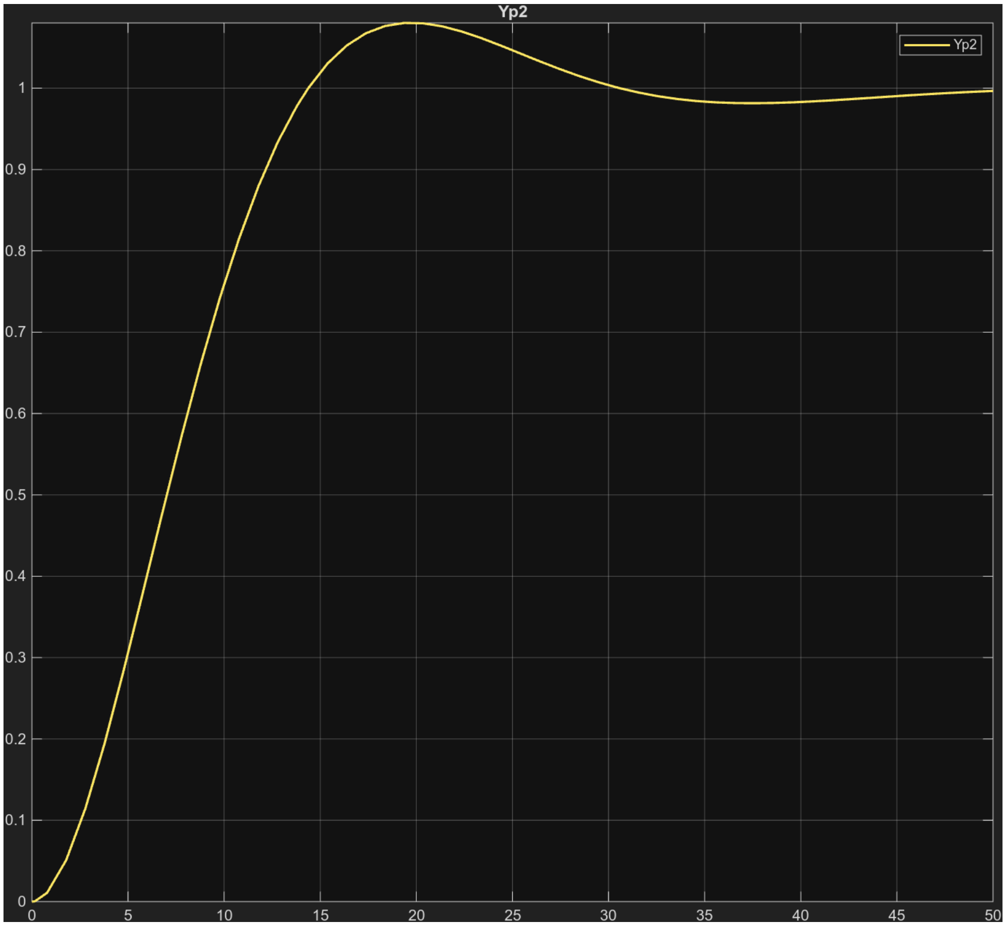

Figure 8: Closed-loop response to unit step setpoint change with no disturbance change from Simulink using PI autotuning.

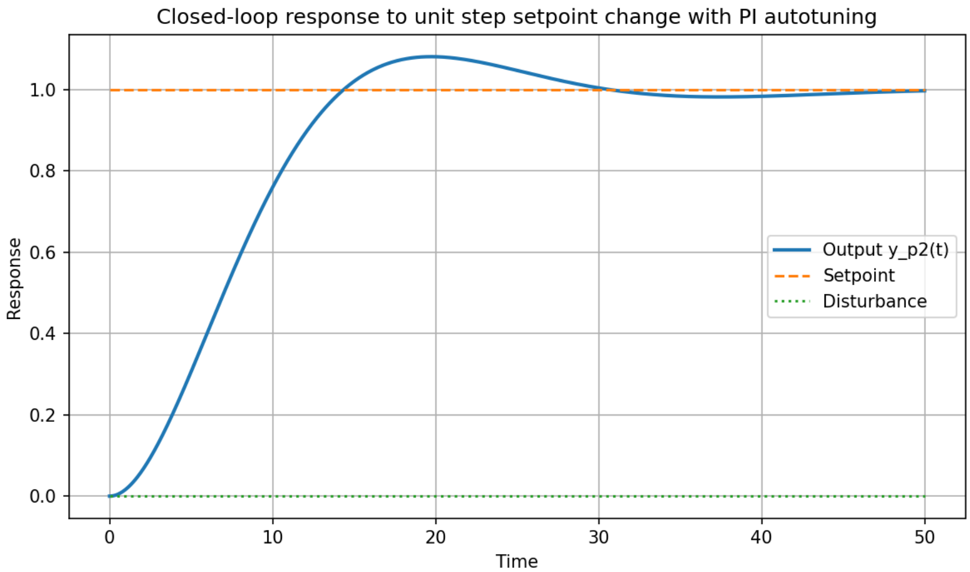

Figure 9: Closed-loop response to unit step setpoint change with no disturbance change from Python using PI autotuning.

The response to a unit step change in setpoint with no disturbance change using PI autotuning is shown in Figure 9.